Finite Observation and Broader Physical Processes

J. R. Landers

Abstract

A system can pass every test of a simple model over a finite experiment and still belong to larger dynamics that dominate it later. This note makes that statement quantitative. We first bound how well noisy, adaptive measurements can tell two models apart, then build an explicit pair: two coupled Hamiltonian oscillators whose visible coordinate tracks a single isolated oscillator as closely as one likes over a chosen window, yet differs from it by a full amplitude afterwards. The total energy stays bounded, and letting the observer actively drive the system on a finite impulse budget does not break the bound. If the observer records only the sign of position, the two models agree *exactly* until the slow envelope first vanishes. A separate Poisson model—an assumption, not a consequence of the construction—says precisely when further relevant scales exist almost surely. Every bound is proved.

The gap between fitting a record and knowing a system

Suppose you watch a system for a while and a simple model fits: residuals sit at the noise level and predictions hold up. What does that license? It licenses prediction over the range you tested. It does not by itself rule out that the system is one part of larger dynamics whose remaining parts are quiet during your record and loud afterwards. This note measures the gap between the two claims.

A model can fit a finite record arbitrarily well and still omit dynamics that dominate the system later.

Fit arbitrarily well has a precise meaning here: no protocol in a stated class—fixed noise level, fixed number of readings, adaptive sampling, bounded driving—distinguishes the two models better than chance by more than ε. *Larger* means having a characteristic timescale longer than the record; the hidden dynamics need not sit anywhere else in space. In our example everything happens inside a two-degree-of-freedom mechanical system that exchanges energy slowly.

The argument runs in five steps. A geometric estimate asks how much of a slow variation fits inside a short window. A statistical bound turns small signal differences into a limit on detection. The oscillator pair then supplies a difference that is invisible early and total later, and a second bound shows that active probing on a finite budget does not help. A coarse-graining hides the difference exactly rather than approximately. A final, separate section asks a different question: how likely is a further scale in the first place?

The tools are standard graduate material—Taylor remainders, Gaussian likelihood ratios, normal modes, Poisson point processes—and what they buy is a clean separation between predictive reliability and physical completeness. Noise is essential to the story: with noiseless data, spectral methods can resolve arbitrarily close frequencies from a single snapshot [1].

What fits inside an observation window

Begin with the simplest version of the question, one with no dynamics in it at all. The observer has access to a closed ball $B_R(x_0) \subset \mathbb{R}^d$ carrying a Euclidean metric, and measures a scalar observable f, differentiable near that ball. The radius R is the reach of the record—how far the measurements extend—not the size or lifetime of the observer. If the ball mixes spatial and temporal coordinates, they must first be put into common units by an explicit physical normalization.

Fix an amplitude scale $A > 0$ and a variation scale $L > 0$, and suppose $\|\nabla f\| \le A/L$ on the ball. This inequality does all the work below, and we assume it rather than derive it: saying that a process has scale L does not by itself imply it.

Proposition 1. Local variation and curvature. If $\|\nabla f(x)\| \le A/L$ throughout $B_R(x_0)$, then

$$\sup_{x \in B_R(x_0)} |f(x) - f(x_0)| \le A\frac{R}{L}. \tag{1}$$

If instead f is twice continuously differentiable and $\left\|\nabla^2 f(x)\right\|_{\mathrm{op}} \le A/L^2$, then its affine approximation $\ell(x) = f(x_0) + \nabla f(x_0) \cdot (x - x_0)$ obeys

$$\sup_{x \in B_R(x_0)} |f(x) - \ell(x)| \le \frac{A}{2}\left(\frac{R}{L}\right)^2. \tag{2}$$

Proof. Write $h = x - x_0$ and $\gamma(u) = x_0 + uh$, $0 \le u \le 1$. The segment stays inside the ball. The fundamental theorem of calculus gives

$$f(x) - f(x_0) = \int_0^1 \nabla f(\gamma(u)) \cdot h \, du,$$

whose absolute value is at most $A\|h\|/L \le AR/L$. For the second statement, applying the same theorem twice gives the integral remainder

$$f(x) - \ell(x) = \int_0^1 (1-u) h^{\mathsf{T}} \nabla^2 f(\gamma(u)) h \, du.$$

Its absolute value is at most $(A/L^2)\|h\|^2 \int_0^1 (1-u)\,du \le AR^2/(2L^2)$. □

Read the two bounds as statements about what a record can look like. When $R \ll L$, the first says the observable is constant to within AR/L: a slow process is indistinguishable from a fixed value. The second says that even after fitting value and slope, the leftover curvature is only $A(R/L)^2/2$, so the process looks like a simple trend. Neither bound claims the dynamics are hidden in principle—a different observable of the same system may show the variation at once.

From a small signal difference to a limit on detection

A difference of size b between two signals is not yet a difference an experiment can detect. That depends on the noise, on how many readings are taken, and on what the observer may do between them. We fix all of this in a protocol $\mathcal{O}$: the accessible coordinates, the observation window, the measurement noise, the sampling policy (which may be adaptive), and any forcing permitted. Each physical model then induces a law $P^{\mathcal{O}}$ on the complete record—every reading, in order, together with the actions taken. Define

$$\mathrm{TV}(P,Q) = \sup_B |P(B) - Q(B)|, \qquad D_{\mathrm{KL}}(P\|Q) = \int \log\left(\frac{dP}{dQ}\right) dP,$$

where relative entropy uses natural logarithms and is infinite when absolute continuity fails. Equal record laws mean the protocol cannot distinguish the models at all; we call two laws ε-*unresolved* when their total variation is at most ε. This approximate relation need not be transitive, so ε-unresolvedness should not be chained.

Lemma 2. A bound for adaptive Gaussian records. Consider two models with at most n scalar readings. Given the same preceding record, reading i has respective conditional means $m_{0,i}, m_{1,i}$ and common variance $\sigma^2 > 0$. Assume $|m_{1,i} - m_{0,i}| \le b$ for every allowed preceding record and action. Policies use only preceding readings and independent randomization. Then

$$D_{\mathrm{KL}}(P_1^{\mathcal{O}}\|P_0^{\mathcal{O}}) \le \frac{nb^2}{2\sigma^2}, \qquad \mathrm{TV}(P_1^{\mathcal{O}}, P_0^{\mathcal{O}}) \le \min\left\{1, \frac{\sqrt{n}\,b}{2\sigma}\right\}. \tag{3}$$

At equal prior probabilities, the minimum classification error is

$$e_* = \frac{1 - \mathrm{TV}(P_1^{\mathcal{O}}, P_0^{\mathcal{O}})}{2}. \tag{4}$$

Proof. For Gaussian densities of common variance, taking the expectation of their log-density ratio gives $(m_1 - m_0)^2/(2\sigma^2)$. Factor the joint record density into successive conditional densities. The expected log of their product is the sum of the expected conditional log-density ratios, so

$$D_{\mathrm{KL}}(P_1^{\mathcal{O}}\|P_0^{\mathcal{O}}) = \sum_{i=1}^{n} \mathbb{E}_1 \frac{(m_{1,i} - m_{0,i})^2}{2\sigma^2} \le \frac{nb^2}{2\sigma^2}.$$

Conditional on the same history, the policy chooses the same action law in both models; it contributes no additional likelihood ratio. Independent randomization can be included in the record. An early stopping rule is handled by appending model-independent dummy readings after stopping.

For completeness, the variation–entropy inequality used here follows by coarse-graining to an event B. Write $p = P(B)$ and $q = Q(B)$. Jensen's inequality for the density ratio separately on B and B^c gives

$$D_{\mathrm{KL}}(P\|Q) \ge p\log(p/q) + (1-p)\log((1-p)/(1-q)) =: d(p,q).$$

For $0 < p, q < 1$, as a function of p, $d(q,q) = \partial_p d(q,q) = 0$ and $\partial_p^2 d(p,q) = 1/[p(1-p)] \ge 4$. Integrating twice yields $d(p,q) \ge 2(p-q)^2$; boundary cases follow by limits. Taking the supremum over B proves $\mathrm{TV}(P,Q) \le \sqrt{D_{\mathrm{KL}}(P\|Q)/2}$. Finally, a test choosing model 1 on B has error $[P_0(B) + P_1(B^c)]/2 = [1 - (P_1(B) - P_0(B))]/2$. Optimizing over B, or the corresponding likelihood-ratio test, gives (4); randomized tests cannot improve the optimum. □

The content of the lemma is that adaptivity does not help. A clever sampling policy can choose when and where to look, but given the same past record it chooses the same way under either model, so the choice contributes nothing to the likelihood ratio. The observer is left with n readings, each worth at most $b^2/2\sigma^2$ nats.

Combining Proposition 1 and Lemma 2, adaptive samples of f and its constant approximation in $B_R(x_0)$ satisfy

$$\mathrm{TV}(P_f^{\mathcal{O}}, P_{f(x_0)}^{\mathcal{O}}) \le \min\left\{1, \frac{\sqrt{n}A}{2\sigma}\frac{R}{L}\right\}. \tag{5}$$

So $L \ge \sqrt{n}AR/(2\sigma\varepsilon)$ suffices for ε-unresolvedness against the constant model, and comparing against the affine approximation instead replaces the bound by $\sqrt{n}A(R/L)^2/(4\sigma)$. One caveat on the comparison: the approximating value and slope are fixed model parameters, not quantities fitted to the same data. A composite null estimated from the record needs its own treatment.

A hidden degree of freedom that matters later

Indistinguishability on its own is not interesting: two models can be indistinguishable because they barely differ. What we want is a pair that no early experiment separates and whose later behaviour differs completely, with no runaway energy making the difference cheap. Two coupled oscillators do it. All masses are one, and the reference model is a single visible coordinate q with momentum p and Hamiltonian

$$H_0(q,p) = \tfrac{1}{2}p^2 + \tfrac{1}{2}\omega^2 q^2, \qquad \omega > 0. \tag{6}$$

The extension adds a coordinate Q and momentum P:

$$H_\delta(q,p,Q,P) = \tfrac{1}{2}(p^2+P^2) + \tfrac{1}{2}(\omega^2+\delta^2)(q^2+Q^2) + 2\omega\delta qQ, \qquad 0 < \delta \le \omega/2. \tag{7}$$

The observer measures q and, when allowed, drives it; Q, P, and the spring parameters are inaccessible. Note that the extension changes two things at once: it adds the coordinate Q, and it shifts the local stiffness from ω^2 to $\omega^2+\delta^2$. Both are part of what is being proposed.

Proposition 3. Stable dynamics and delayed modulation. For initial data $q(0) = A > 0$, $p(0) = 0$, and $Q(0) = P(0) = 0$ in the extension,

$$q_0(t) = A\cos(\omega t), \qquad q_\delta(t) = A\cos(\omega t)\cos(\delta t), \qquad Q_\delta(t) = -A\sin(\omega t)\sin(\delta t). \tag{8}$$

The extended Hamiltonian is positive definite, and its conserved energy is

$$E_\delta = \tfrac{1}{2}A^2(\omega^2+\delta^2) \le \tfrac{5}{8}A^2\omega^2. \tag{9}$$

Proof. The orthogonal canonical transformation $u = (q+Q)/\sqrt{2}$, $v = (q-Q)/\sqrt{2}$, with momenta $p_u = (p+P)/\sqrt{2}$, $p_v = (p-P)/\sqrt{2}$, gives

$$H_\delta = \tfrac{1}{2}[p_u^2 + (\omega+\delta)^2 u^2] + \tfrac{1}{2}[p_v^2 + (\omega-\delta)^2 v^2].$$

Both frequencies are positive, proving positive definiteness and stability. The initial normal coordinates are $u(0) = v(0) = A/\sqrt{2}$ with zero momenta. Their cosine solutions, transformed back and simplified by the sum and difference identities, give (8). Evaluating the autonomous Hamiltonian at the initial state gives (9). □

The transformation in the proof diagonalizes the extension into two independent oscillators at frequencies $\omega \pm \delta$, and everything follows from that. Both frequencies are real and positive, so the motion is bounded and the energy conserved. Projected back onto the visible coordinate, the two nearby frequencies beat: a carrier at ω underneath an envelope at δ. The hidden process is slow rather than large—timescale $1/\delta$—and it merely moves a fixed amount of energy back and forth.

Theorem 4. Early ambiguity and a later full-amplitude difference. For at most n adaptive position readings on $[0,T]$ with independent Gaussian noise of standard deviation $\sigma > 0$ and no forcing,

$$\mathrm{TV}(P_\delta^{\mathcal{O}}, P_0^{\mathcal{O}}) \le \min\left\{1, \frac{\sqrt{n}A\delta^2T^2}{4\sigma}\right\}. \tag{10}$$

For any positive integer m, choose

$$\delta = \frac{\omega}{4m}, \qquad T_* = \frac{\pi}{2\delta} = \frac{2\pi m}{\omega}. \tag{11}$$

Then $q_0(T_*) = A$ and $q_\delta(T_*) = 0$. For fixed A, ω, T, n, σ, the bound in (10) tends to zero as $m \to \infty$, while this later discrepancy remains exactly A.

Proof. From (8) and $0 \le 1 - \cos z \le z^2/2$,

$$|q_\delta(t) - q_0(t)| \le A[1-\cos(\delta t)] \le A\delta^2T^2/2 \quad (0 \le t \le T).$$

Apply Lemma 2 with $b = A\delta^2T^2/2$. At the selected time, $\omega T_* = 2\pi m$ and $\delta T_* = \pi/2$, proving the exact late values. Finally $\delta^2 = \omega^2/(16m^2) \to 0$. □

This theorem is the core of the note, so read (10) and (11) together. Shrinking δ makes the early record arbitrarily uninformative—the bound falls like $1/m^2$—while the eventual discrepancy stays exactly A, one full amplitude, however small δ is. Only the time at which it appears moves out, to $T_* = 2\pi m/\omega$. There is no tradeoff in which a well-hidden extension must also be a weak one.

Nor is the late effect a small correction riding on the visible motion. At T_* both positions are zero, but $p_\delta(T_*) = -A\delta$ while $P_\delta(T_*) = -A\omega$: the hidden momentum carries the fraction $\omega^2/(\omega^2+\delta^2) \to 1$ of the total energy. A degree of freedom that started with none of the energy ends up with essentially all of it.

Can active probing reveal the extension?

An observer with a controllable system does more than watch it. Pushing on q and seeing how it responds is exactly how one would look for an extra spring, so we allow it. Let a locally integrable force $u(t)$ act on q in both models, subject to a pathwise budget

$$\int_0^T |u(t)|\,dt \le J. \tag{12}$$

on total absolute impulse. The budget is what makes the question non-trivial: an unbounded push separates any two models. Between readings the force may depend on everything recorded so far plus independent randomization, sampling times are nondecreasing and may be adaptive, and the same policy is applied to both models. What the controller may not use is state it has not recorded.

Theorem 5. A uniform bound under bounded probing. For every such policy with at most n position readings of noise level σ,

$$\mathrm{TV}(P_\delta^{\mathcal{O}}, P_0^{\mathcal{O}}) \le \min\left\{1, \frac{\sqrt{n}\,\delta^2}{2\sigma}\left(\frac{AT^2}{2} + \frac{JT^3}{6}\right)\right\}. \tag{13}$$

During probing, each model's instantaneous mechanical energy satisfies

$$H(t) \le \tfrac{1}{2}\left(\sqrt{2H(0)} + J\right)^2. \tag{14}$$

Proof. For a prescribed force history, linearity gives $q_a^u(t) = q_a(t) + \int_0^t G_a(t-s)u(s)\,ds$, where $a = 0, \delta$ and

$$G_0(r) = \frac{\sin(\omega r)}{\omega}, \qquad G_\delta(r) = \frac{1}{2}\left[\frac{\sin((\omega+\delta)r)}{\omega+\delta} + \frac{\sin((\omega-\delta)r)}{\omega-\delta}\right].$$

To bound the difference, write $g_\nu(r) = \int_0^r \cos(\nu s)\,ds$. Differentiating under the integral yields $|\partial_\nu^2 g_\nu(r)| = |\int_0^r -s^2\cos(\nu s)\,ds| \le r^3/3$. Taylor's formula about $\nu = \omega$, averaged at $\omega \pm \delta$, cancels the linear terms and gives

$$|G_\delta(r) - G_0(r)| \le \delta^2 r^3/6.$$

Consequently, for $t \le T$ and every force satisfying (12),

$$|q_\delta^u(t) - q_0^u(t)| \le \delta^2\left(\frac{AT^2}{2} + \frac{JT^3}{6}\right) =: b_J. \tag{15}$$

In an adaptive experiment, fix a common preceding record and random seed. The policy has then generated the same force history and next sampling time under either model. Thus (15) bounds each conditional mean difference, even though force histories on two separately realized runs need not coincide. Lemma 2 proves (13).

For energy, Hamilton's equations with the external force give $\dot H = pu$. Positive definiteness implies $|p| \le \sqrt{2H}$. For $a > 0$, $d\sqrt{2H + a}/dt = pu/\sqrt{2H+a} \le |u|$ almost everywhere. Integrating, then taking $a \downarrow 0$, proves (14). □

The forced bound (13) carries the same δ^2 prefactor as the passive one, with the impulse budget contributing a term of order JT^3. Driving the system helps, but only by a fixed factor; it cannot beat the $\delta^2 \to 0$ scaling. Inequality (14) confirms that the observer is not smuggling in unlimited resources: on an impulse budget J the energy stays bounded whatever the force does.

Corollary 6. No uniform finite-budget certificate of closure. Fix $A, \omega, T, \sigma > 0$, a finite n, $J \ge 0$, and $0 < \varepsilon < 1$. There is a nonzero $\delta = \omega/(4m)$ such that *every* allowed policy has $\mathrm{TV}(P_\delta^{\mathcal{O}}, P_0^{\mathcal{O}}) \le \varepsilon$, yet the unforced models differ by A at a later time $T_* > T$. Their unforced energies and energies during allowed probing are bounded uniformly in m.

Proof. Choose an integer m large enough that $2\pi m/\omega > T$ and

$$\frac{\sqrt{n}\,\omega^2}{32\sigma m^2}\left(\frac{AT^2}{2} + \frac{JT^3}{6}\right) \le \varepsilon.$$

Theorem 5 applies uniformly over all permitted policies. Theorem 4 supplies the unforced late difference, and (9)–(14) supply the uniform energy bounds. □

Read the corollary narrowly. It says that no fixed budget—n, σ, T, J—certifies the absence of extensions *uniformly* over a family containing arbitrarily weak nonzero coupling. It does not say that any given extension is undetectable: fix δ, and a longer window, a finer instrument, another observable, or a harder push will find it. The full-amplitude late difference is also a statement about the unforced trajectories; after an intervention the models still differ, but we do not claim they differ by exactly A.

For passive observation the same inequalities price the alternative strategy of waiting. Requiring classification error at most $\alpha < 1/2$, (4) and (10) require

$$n \ge \frac{16\sigma^2(1-2\alpha)^2}{A^2\delta^4T^4} = \frac{256\sigma^2(1-2\alpha)^2}{\pi^4A^2}\left(\frac{T_*}{T}\right)^4. \tag{16}$$

The sample count grows as the fourth power of T_*/T: reaching a consequence a hundred times further out than the record costs on the order of 10^8 times the samples. This is a lower bound rather than an achievability claim, but the tradeoff it makes explicit is the practical one—at fixed amplitude and noise, buying access to a much later consequence from an early record gets expensive fast.

A coarse record that hides the change exactly

The concealment so far is approximate: the records differ, just by too little to resolve. Coarse-graining can make it exact. Suppose the observer keeps only the sign of position:

$$\Pi(q) = \begin{cases} +, & q \geq 0, \\ -, & q < 0. \end{cases}$$

This partition sends an entire half-line to a single symbol. The extension changes the amplitude of the visible oscillation, and a change in positive amplitude moves *within* a cell rather than across its boundary—so it leaves no trace at all in the symbolic record until the envelope itself changes sign.

Proposition 7. Exact symbolic agreement before the envelope zero. For the unforced solutions, at every $0 \leq t < \pi/(2\delta)$,

$$\Pi(q_\delta(t)) = \Pi(q_0(t)). \tag{17}$$

For the discrete choices in (11), at $2T_*$ one has $q_0(2T_*) = A$ and $q_\delta(2T_*) = -A$, so the symbols differ.

Proof. Before $\pi/(2\delta)$ the factor $\cos(\delta t)$ is strictly positive. Multiplication by it preserves sign, including common zeros under the stated convention. At $2T_*$, $\omega(2T_*) = 4\pi m$ and $\delta(2T_*) = \pi$, giving the asserted values. □

Two caveats. This is an idealized sign-only channel: an observer who instead thresholds a noisy amplitude does not get identical record laws, because the amplitude changes the probability of a sign error—for that case the earlier statistical bound applies. And a symbolic record that looks regular is not thereby Markov; the unresolved phase can still control the next transition.

The connection to finite-horizon coarse descriptions is precise. In the continuous-computation framework [2, 3] one compares a decoded symbolic path with a finite-state reference process over a stated epoch. The next statement transfers any *independently established* approximation bound from one model to the other; it assumes nothing about lumpability of the oscillator's sign partition.

Proposition 8. Transfer of a finite-horizon predictive description. Let a common decoder, possibly randomized, map experimental records to symbolic paths, with laws S_0 and S_1. If $\mathrm{TV}(P_1^{\mathcal{O}}, P_0^{\mathcal{O}}) \leq \varepsilon$ and a reference path law M satisfies $\mathrm{TV}(S_0, M) \leq e$, then

$$\mathrm{TV}(S_1, M) \leq \min\{1, e + \varepsilon\}. \tag{18}$$

Proof. Let $K(B \mid r) \in [0,1]$ be the decoder's probability of a symbolic event B given record r. Then $S_1(B) - S_0(B) = \int K(B \mid r)\, d(P_1^{\mathcal{O}} - P_0^{\mathcal{O}})(r)$. For any measurable $h \in [0,1]$, write $h(r) = \int_0^1 \mathbf{1}\{h(r) > s\}\, ds$; integration shows $|\int h\, d(P_1^{\mathcal{O}} - P_0^{\mathcal{O}})| \leq \mathrm{TV}(P_1^{\mathcal{O}}, P_0^{\mathcal{O}})$. Hence $\mathrm{TV}(S_1, S_0) \leq \varepsilon$. The triangle inequality gives the result. □

A model that is wrong about the physics can therefore inherit a certified finite-horizon predictive description from the correct one, paying only ε in extra error. The certificate covers the stated preparation and record class, not worst-case behaviour over all hidden states, and the horizon a given error budget certifies is a lower bound on how long the description works, not the point where it fails.

A worked example with numbers

Take $A = \omega = 1$, $\delta = 1/400$, $T = 10$, $n = 100$, and $\sigma = 0.1$, so that $m = 100$ and $T_* = 200\pi \approx 628.319$. The analytic bounds give

$$\sup_{0 \leq t \leq 10} |q_\delta(t) - q_0(t)| \leq 0.0003125, \qquad \mathrm{TV} \leq 0.015625, \qquad e_* \geq 0.4921875.$$

The best possible equal-prior classifier therefore succeeds at most 50.78125% of the time on this protocol class—barely better than a coin flip. Yet at T_* the two models' position predictions differ by exactly one full amplitude, and by $2T_*$ even their signs disagree.

An independent integration of Hamilton's equations with relative tolerance 10^{-10} and absolute tolerance 10^{-12} agreed with the exact visible trajectory to within 3×10^{-9} on $[0, T_*]$, with absolute energy drift below 8×10^{-9}. For the test force $u(t) = 0.1 \sin(0.7t)$ on $[0, 10]$, the absolute impulse is at most $J = 1$. The maximum computed position difference was 0.00024649, below the uniform bound $b_J = 0.00135417$. The forced record bound is then $\mathrm{TV} \leq 0.067709$ for the same n, σ. These runs check the implementation only; the inequalities above are proved and do not rest on a fit to simulation.

How likely is a further scale in the first place?

The oscillator construction shows that consequential hidden extensions exist. It says nothing about how often they occur, and for that one needs a population model—an assumption rather than a result. Read this section as the price of the stronger claim: here is exactly what must be assumed for further relevant scales to be certain.

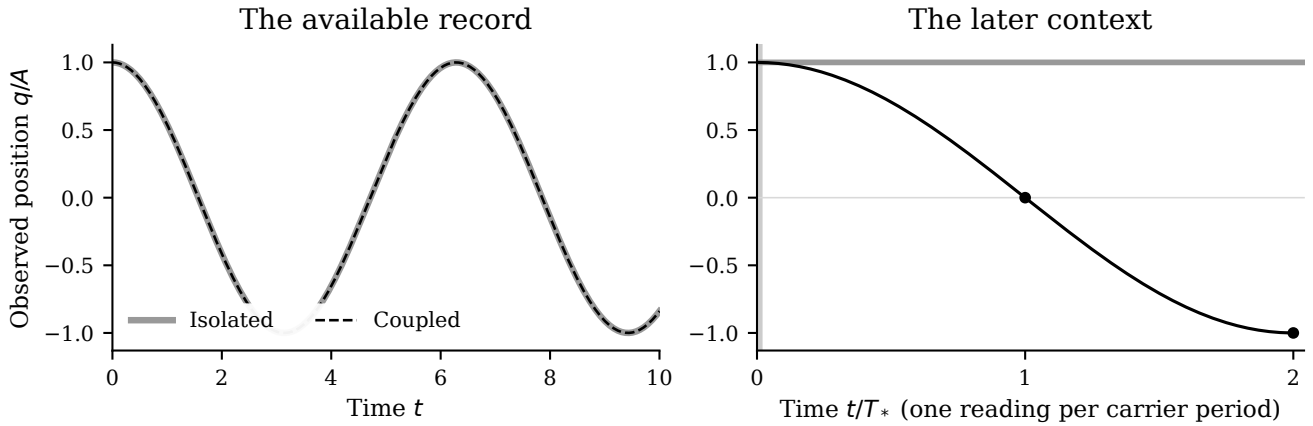


Figure 1: One record, two physical explanations. Left: over the initial window, the isolated and coupled positions nearly coincide. Right: sampling once per carrier period makes the slow envelope visible; the isolated prediction remains one while the coupled prediction eventually reaches zero and then minus one. The narrow shaded interval marks the initial window. All plotted values are exact solutions; connecting lines on the right guide the eye between samples.

Index processes by a characteristic scale $L \geq L_{\min} > 0$. If scale corresponds to a d-dimensional volume $V(L) = v_d L^d$, then

$$\frac{dV}{V} = d\,\frac{dL}{L}.$$

so equal multiplicative steps in scale are equal steps of log volume. This is a choice of coordinate; it asserts nothing about where processes actually sit.

Assume the scales of relevant processes form a Poisson point process with locally finite intensity measure

$$d\mu(L) = \eta(L)\frac{dV}{V} = d\,\eta(L)\frac{dL}{L}, \qquad \eta(L) \geq 0. \tag{19}$$

Relevance here is stipulated, and the stipulation does real work: a counted process must participate in the dynamics or explanation of the phenomenon. Without it the claim would be empty, since one can always point to more distant unrelated systems. Constant η recovers the familiar scale-invariant Poisson process [4].

Theorem 9. A criterion for unbounded process scales. Under (19), the probability of at least one process in $(R, S]$ is

$$\Pr\{N(R,S) \geq 1\} = 1 - \exp\left[-d\int_R^S \frac{\eta(L)}{L}\,dL\right]. \tag{20}$$

For finite $R \geq L_{\min}$ this tends to one as $S \to \infty$ if and only if

$$\int_R^\infty \frac{\eta(L)}{L}\,dL = \infty. \tag{21}$$

If this condition holds, process scales are unbounded almost surely.

Proof. Poisson counts have zero-count probability $e^{-\mu((R,S])}$, which directly gives (20). Equivalently, the survival probability $Q_R(S)$ solves

$$Q_R'(S) = -\frac{d\,\eta(S)}{S}Q_R(S), \qquad Q_R(R) = 1,$$

almost everywhere, whose integrated solution is the exponential in (20). The exponent tends to minus infinity exactly under (21). For $R_k = 2^k L_{\min}$, each event that some point occurs beyond R_k consequently has probability one. Their countable intersection also has probability one, and on that intersection the scales are unbounded. □

The criterion cuts both ways. Constant intensity $\eta_0 > 0$ gives $1 - (R/S)^{d\eta_0} \to 1$, so further scales are certain. But an intensity decaying as a power, $\eta(L) = \eta_0(L_{\min}/L)^a$ with $a > 0$, leaves a positive probability $\exp[-(d\eta_0/a)(L_{\min}/R)^a]$ that nothing lies beyond R at all. The mere fact that observation stops somewhere does not make continuation likely; the intensity law has to supply the weight at large scales, and no finite record establishes which law holds.

Independence is not what drives the conclusion. Suppose scale bands with endpoints tending to infinity have events E_j and, for every $m \leq j$,

$$\Pr(E_j \mid E_m^c \cap \cdots \cap E_{j-1}^c) \geq q > 0$$

whenever the conditioning event has positive probability, with empty conditioning interpreted unconditionally. The probability chain rule gives $\Pr(E_m^c \cap \cdots \cap E_n^c) \le (1-q)^{n-m+1} \to 0$. Taking a countable intersection over m proves that infinitely many E_j occur almost surely. Replacing independence with a uniform conditional continuation assumption removes one hypothesis and installs another; it does not derive continuation from nothing.

Two limits are worth stating explicitly. If an unbounded population also satisfies the derivative bound of Proposition 1 with a common A, then for fixed $R, n, \sigma, \varepsilon > 0$ infinitely many of its processes exceed $\sqrt{n}AR/(2\sigma\varepsilon)$, and each is individually ε-unresolved against its local constant approximation. That says nothing about their combined signal: superposition, mutual coupling, and total energy would need a separate construction. And unbounded scales need not be nested, each process containing the last. Counting points gives neither property.

What the construction shows

The construction is deliberately small: one visible oscillator, one hidden one, bounded energy throughout. Within a stated class of noisy measurements and bounded interventions, its early record approaches that of an isolated oscillator arbitrarily closely, while the hidden coordinate produces a full-amplitude effect later and ends up holding nearly all the energy. Under a sign-only record the concealment is exact rather than approximate.

The moral is a distinction between two tasks that are easy to conflate. Prediction over the tested scope is what a good fit certifies, and that certificate is real. Certifying that no further relevant degrees of freedom exist is a stronger claim, and no fixed budget establishes it uniformly over the family considered here. That last statement is about uniformity: any particular extension yields to a longer window, a better instrument, a different observable, or a harder push.

The probability question is separate, and its answer is conditional. Further relevant scales are almost certain exactly when the integrated intensity diverges, and no finite record establishes that the intensity diverges. The natural next step is to build families in which the scale distribution, couplings, and observation costs are derived from physics together rather than assumed one at a time. In the meantime the construction supplies a working standard for what should count as a serious claim of hidden context: it must be dynamically coupled, hard to identify under stated resources, and consequential at a stated later time.

References

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